

HOST_POWER_MANAGEMENT

Host Power Management — Hard Ban (CONST-033)

Why this exists

On 2026-04-26 18:23:43 the host running mission-critical parallel CLI agents and container workloads was auto-suspended mid-session. This killed the running HelixAgent binary, all 41 dependent services, every SSH session, and every active CLI agent on the box. journalctl showed:

```
systemd-logind[1183]: The system will suspend now!
```

The user-level GNOME power settings were already correct (sleep-inactive-ac-type=nothing). The trigger was the **GDM greeter session at the local console**, which has its own power policy and does not count SSH activity. Earlier, on multiple occasions, the user@1000.service had been SIGKILLED by systemd because heavy memory pressure prevented gnome-shell from responding to GDM/Wayland watchdog within TimeoutStopSec — perceived by the user as "the system fully logged me out." Together these two failure modes have caused repeated loss of in-flight agent work.

The rule

No project shipped from this workspace may invoke a host-level power-state transition. Forbidden invocations include — but are not limited to:

Layer	Forbidden invocations
systemd CLI	systemctl suspend, systemctl hibernate, systemctl hybrid-sleep, systemctl suspend-then-hibernate, systemctl poweroff, systemctl halt, systemctl reboot, systemctl kexec
logind CLI	loginctl suspend, loginctl hibernate, loginctl hybrid-sleep, loginctl suspend-then-hibernate, loginctl poweroff, loginctl halt, loginctl reboot
Legacy CLI	pm-suspend, pm-hibernate, pm-suspend-hybrid, shutdown -h/-r/-P/-H/now, bare reboot/poweroff/halt
DBus	org.freedesktop.login1.Manager. {Suspend,Hibernate,HybridSleep,SuspendThenHibernate,PowerOff,Reboot}, org.freedesktop.UPower.{Suspend,Hibernate,HybridSleep} (via dbus-send, busctl, or any language binding)
gsettings	gsettings set ... sleep-inactive-{ac,battery}-type to anything other than 'nothing' or 'blank'

Defence in depth

Three layers, in order of strength:

- 1. Host-level masking (manual prereq, sudo required, run once):** scripts/host-power-management/install-host-suspend-guard.sh masks sleep.target, suspend.target, hibernate.target, hybrid-sleep.target, writes /etc/systemd/sleep.conf.d/00-no-suspend.conf with AllowSuspend=no, and writes /etc/systemd/logind.conf.d/00-no-idle-suspend.conf with IdleAction=ignore and HandleLidSwitch=ignore. After this, no user / session / DE / greeter / cron job can suspend the host.
- 2. User-session bootstrap (no sudo):** scripts/host-power-management/user_session_no_suspend_bootstrap.sh runs gsettings, xset -dpms, and (opt-in via HOST_POWER_MANAGEMENT_SESSION_INHIBIT=1) systemd-inhibit to protect the current GUI/CLI session of the invoking user. Idempotent. Safe to source from start.sh / setup.sh / bootstrap.sh.
- 3. Source-tree static gate:** scripts/host-power-management/check-no-suspend-calls.sh walks the tree and exits non-zero on any forbidden invocation. challenges/scripts/no_suspend_calls_challenge.sh wraps it as a challenge that runs in CI / run_all_challenges.sh. challenges/scripts/host_no_auto_suspend_challenge.sh asserts the running host's state matches the layer-1 masking.

Verification

```
# Layer 1: host state
bash challenges/scripts/host_no_auto_suspend_challenge.sh
# Expected: 4 PASS
```

```
# Layer 3: source tree
bash challenges/scripts/no_suspend_calls_challenge.sh
# Expected: PASS, no forbidden calls
```

If you genuinely need a power-state transition

You don't, in this workspace. If a future legitimate use case arises (e.g. a container's *internal* init script that suspends a *guest VM*, not the host), add the specific file path to `EXCLUDE_PATHS` at the top of `check-no-suspend-calls.sh` with a comment explaining the non-host context.